Health Charity Donor Intelligence & Fundraising Strategy - Canada

Data Analytics Portfolio Project Brief

Dataset	Statistics Canada - 2023 Survey on Giving, Volunteering and Participating (SGVP) PUMF
Project focus	Donor intelligence, health-charity giving, fundraising channels, motivations, barriers, and actionable targeting opportunities
Core stack	Python / pandas + SQL + statistical analysis + Power BI
Optional extension	Health Donor Propensity analysis (machine learning)

1. Project Objective

The purpose of this project is to use Canadian survey data to understand charitable donors, identify the characteristics of people who support health-related organizations, examine how and why they give, identify barriers to additional giving, and translate those findings into fundraising opportunities for a Canadian health charity.

Primary business question

How can a Canadian health charity identify, understand, and prioritize donor audiences to improve fundraising strategy?

The project is designed as a data storytelling case study. The analysis should not begin with predetermined conclusions. Each stage should answer a business question and use the result to guide the next stage of investigation.

2. Storytelling Framework

THE CANADIAN DONOR

Who gives and how much?

Establish the Canadian giving landscape. Measure donor participation, donation value and frequency, then examine how giving differs across demographic and socioeconomic groups.

Key analytical areas: Donor rate; total giving; median/average donation; donation frequency; age; income; education; province.

THE HEALTH DONOR

What makes health donors different?

Narrow the analysis from all charitable donors to people who financially support health organizations. Compare health donors with other donors and non-donors.

Key analytical areas: Health donor rate; health donation amount; health donation frequency; demographic and socioeconomic profile; volunteering.

HOW THEY GIVE

Which channels and behaviours matter?

Investigate the ways donors respond to fundraising requests and how donor behaviour differs across audience segments.

Key analytical areas: Online requests; telephone; traditional media; door-to-door; checkout/street; sponsored events; donor loyalty; charity research behaviour.

WHY THEY GIVE

What motivates them?

Examine motivations behind charitable giving and determine whether health donors show distinctive motivational patterns.

Key analytical areas: Personal connection to the cause; belief in the cause; compassion; community contribution; social influence; tax motivation.

WHY THEY DON'T GIVE MORE

What are the barriers?

Identify the factors that prevent donors from increasing their financial support and distinguish financial constraints from trust, awareness, and relevance barriers.

Key analytical areas: Affordability; satisfaction with current giving; difficulty finding causes; preference for volunteering; efficiency concerns; fundraising-cost perceptions.

THE OPPORTUNITY

Which audiences and channels should a health charity prioritize?

Synthesize the evidence from the previous stages into a small number of defensible fundraising opportunities. Recommendations must be explicitly connected to observed data.

Key analytical areas: Priority audiences; appropriate channels; relevant motivations; barriers to address; evidence supporting each recommendation.

3. Analytical Workflow

Stage	Purpose	Main output
1. Business understanding	Define the fundraising questions before touching the data.	Project questions and scope
2. Data understanding	Review the SGVP structure, codebook, variable meanings, survey weights and special response codes.	Data dictionary / variable map
3. Data quality assessment	Profile distributions, valid skips, refusals, missing-like codes, outliers and applicability rules.	Data quality report
4. Data preparation	Create a reproducible analytical dataset containing only required variables and documented transformations.	Clean analysis table
5. Exploratory analysis	Answer the six storytelling chapters progressively, using weighted estimates where appropriate.	Analytical tables and findings
6. SQL layer	Create reproducible queries/views for aggregations used in reporting.	SQL scripts / analytical views
7. Power BI	Turn validated findings into an executive narrative rather than a collection of unrelated charts.	Interactive dashboard
8. Recommendations	Connect evidence to practical fundraising decisions.	3-5 evidence-based opportunities
9. Optional ML	Explore characteristics associated with health-donor propensity without making ML the centre of the project.	Bonus propensity analysis

4. Tool Roles

Python / pandas: data profiling, cleaning, validation, exploratory analysis, statistical comparisons, feature creation, and optional modelling.

SQL: reproducible transformations and aggregations, analytical views, donor segment summaries, and Power BI-ready tables.

Power BI: final storytelling layer. It should communicate the findings to a fundraising or management audience and make the progression from landscape to opportunity easy to follow.

Machine learning: optional extension only. If included, it should answer a clear question such as which observable characteristics are associated with health-charity giving. It should not replace the descriptive and diagnostic analysis.

5. Proposed Power BI Story

Page	Purpose
01 - Canadian Giving Landscape	Who gives, how much, and how frequently? Establish the baseline.
02 - The Health Donor	Show how health donors differ from the broader donor population.
03 - Channels & Behaviour	Explain how different donor audiences give and interact with charities.
04 - Motivations & Barriers	Show what drives giving and what prevents additional giving.
05 - Fundraising Opportunities	Summarize priority audiences, channels, and evidence-based recommendations.

6. Important Methodological Guardrails

Use survey weights: SGVP is survey microdata. Population-level percentages and estimates should use the person weight rather than treating raw respondent counts as population totals.

Do not assume special codes are missing: valid skip, don't know, refusal and not stated codes vary by variable. Cleaning rules must be based on the official codebook.

Do not imply causality: observed differences and associations should be described as patterns in the survey data unless the analysis design supports stronger claims.

Do not present this as Canadian Cancer Society CRM data: the project uses public Canadian survey data as market and donor intelligence for a hypothetical health-charity fundraising context.

Recommendations must follow the evidence: audience and channel recommendations should be written only after the analysis establishes the relevant patterns.

7. Dataset

Statistics Canada - 2023 Survey on Giving, Volunteering and Participating (SGVP), Public Use Microdata File.

8. Definition of Success

A successful final project should allow a reviewer to understand, within a few minutes, who Canadian donors are, how health donors differ, how they give, what motivates or limits their giving, and which fundraising opportunities are supported by the evidence. Technical work should be reproducible, but the final deliverable should prioritize clarity, business relevance, and data storytelling.